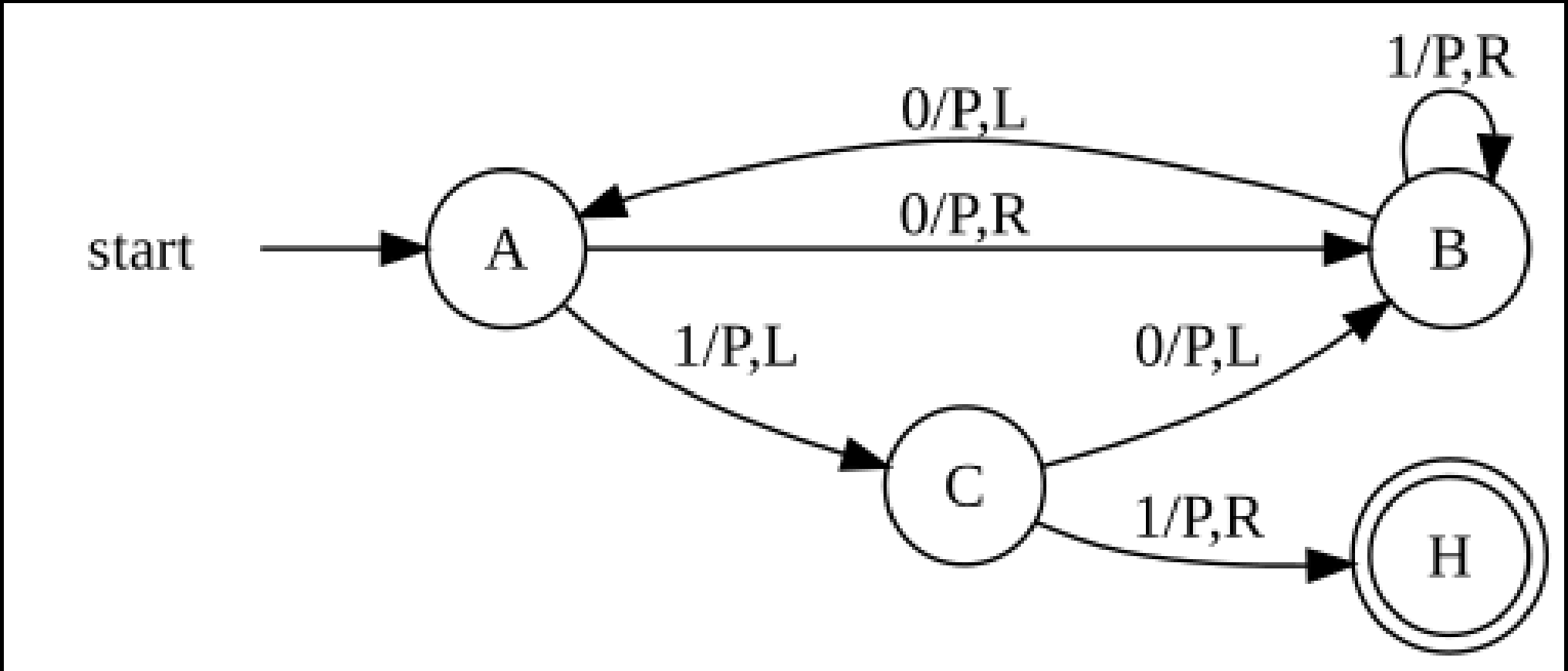
STATE MACHINE

- General state machines.
- Relating to games.

TURING MACHINE



STATE MACHINE

- Each state contains:
  - Possible actions.
  - Observations.
  - Exit to other states.
- Put more simply:
- What am I doing?
  - Update position, take an action.
- What is going on?
  - Where are the other actors, obstacles, goals.
- Should I be doing something else?
  - Is something in my variables triggering a transition.

MARIO

- Mario has four states:
- Running slow.
  - Slow or stopped.
  - Can change to any direction immediately.
- Running fast.
  - Faster speed
  - Must reduce to **Running Slow** to turn.
- Diving.
  - Only state in which Mips can be caught.
- Standing up.
  - Cost of diving is remaining stopped briefly, transitions back to running slow.

RUNNING SLOW

- Observation:
  - Direction and magnitude of input.
  - Dive button
- Action:
  - Continue moving in current direction of the input, modifying speed based on magnitude of input.
- Transition:
  - Speed increased above fast threshold. (To Fast)
  - Dive requested. (To Dive)

RUNNING FAST

- Observation:
  - Direction and magnitude of input.
  - Dive button
- Action:
  - Move in direction of input if within angle tolerance.
  - Slow down otherwise.
- Transition:
  - Speed decreased below fast threshold. (To Slow)
  - Dive requested. (To Dive)

DIVING

- Observation:
  - Time elapsed since dive started
  - Rabbit within reach
- Action:
  - Move in direction of dive.
  - Attach rabbit to Mario if within reach.
- Transition:
  - Dive time reached. (To Standup)

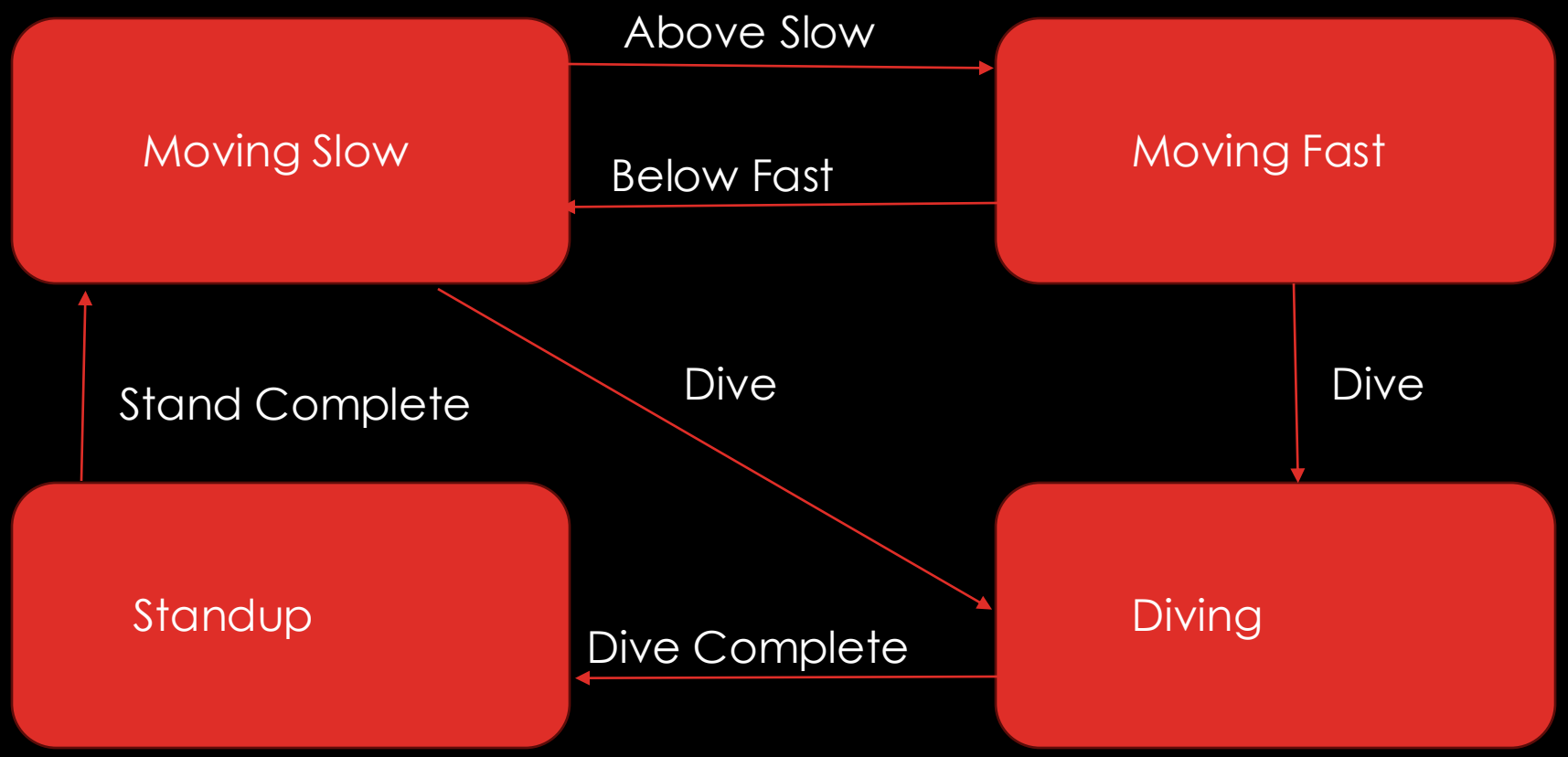
- Note that input is ignored during the dive.

STANDUP

- Observation:
  - Time elapsed since standup started
- Action:
  - Don't move.
- Transition:
  - Standup time reached. (To Move Slow)

- Input is also ignored during standup.

PUTTING IT ALL TOGETHER



HELPER FUNCTIONS

- Direction of movement.
  - The raw direction requested.
- Magnitude of movement.
  - How fast is the requested speed (distance of mouse from player, simulating a joystick.
- Distance between current direction and input angle.
  - Used for fast speed determination if player must slow down before turning.
- Dive button.
  - Get the bunny!

STATES VS VARIABLES

- States are best when an action will lead to a distinct change in behavior.
- Most useful for our understanding as programmers.
  - Even if states seem similar, separating them allows easier debugging/unit testing.
- Variables track actual properties of the entity.
- States watch variables as a method of determining when to transition to different states.
- Variables we wouldn't want as states in and of themselves due to too many possibilities:
  - Transform (position/rotation/scale)
  - Speed
  - Direction of movement
- Alternatively, a couple things that make individual states over complicated (add another state):
  - "Boolean stacking", ie: using booleans to track doing different things in a single state. This is why moving fast and slow are separate states rather than a boolean in one state. While even one boolean in a state is "dangerous", if you find you are using two or more to control behavior that is a "code smell" that senior programmers will detect.
  - Transitioning to too many other states. This indicates there should be an intermediate state.